ESWCap C+R Meter, carrier board Rev C.1: build guide

Team Praise Claude · Embedded Systems Workshop · generated 2026-11-14 from the verified design (`hardware/kicad/Project.kicad_sch` + `Project.kicad_pcb`, schematic ↔ PCB parity clean). 88 parts, 70 nets, 124 point-to-point wires + 82 rail connections.

Files that come with this guide: **ESWCap-wiring-map.drawio** (editable, one layer per block), **ESWCap-wiring-map.pdf** (printable), **ESWCap-wire-list.csv** (spreadsheet checklist), and the circuit diagrams in `docs/circuit/ESWCap-circuit-diagrams.pdf`.

STOP — read before powering anything.
1. **Firmware:** the current firmware drives PB2 HIGH and PB15 LOW at boot. On Rev C.1, PB2 is the output of U1B (VBUF) and PB15 is the R-mux enable. **Do not plug a programmed WeAct module into the assembled board until the firmware pin-map update (docs/accuracy-plan.md, item F1) is flashed.** Tests 1–4 below are done without the module.
2. **V_BIAS:** leave J6 disconnected until every test passes. Then start at 5 V, never above 20 V.
3. **USB power only.** The relays draw ≈ 50 mA each from the module's 3.3 V regulator.

1. Parts

References	Value / part	Footprint	Qty
C1	1000uF/35V BP	C_Radial_D16_P7.5	1
C2	68nF	C_0805_2012Metric	1
C3, C20	100pF	C_0805_2012Metric	2
C4, C21	10pF	C_0805_2012Metric	2
C5, C6, C8, C10, C11, C12, C13, C14, C17	100nF	C_0805_2012Metric	9
C7	1nF	C_0805_2012Metric	1
C9, C15, C16	10uF	C_0805_2012Metric	3
D1, D2	BAV199	SOT-23	2
D3	BAT54S	SOT-23	1
D4, D5, D6	1N4148	D_SOD-123	3
J1	WeAct P1	PinSocket_1x20_P2.54mm_Vertical	1
J2	WeAct P2	PinSocket_1x20_P2.54mm_Vertical	1
J3	WeAct P3 (NC)	PinSocket_1x04_P2.54mm_Vertical	1
J4	WeAct P4 3V3	PinSocket_1x04_P2.54mm_Vertical	1
J5	DUT	TerminalBlock_4P_5.08	1
J6	V_BIAS 0-20V	TerminalBlock_2P_5.08	1
J7	Console	PinSocket_1x04_P2.54mm_Vertical	1
J8	BIAS CTRL	PinHeader_1x08_P2.54mm_Vertica	1

K1	R_MODE	Relay_DPDT_Omron_G6S-2	1
K2	KELVIN	Relay_DPDT_Omron_G6S-2	1
Q1	AO3401	SOT-23	1
Q2, Q3	BC847	SOT-23	2
R1	98.9 1%	R_0805_2012Metric	1
R2	993.6 1%	R_0805_2012Metric	1
R3	98.6k 1%	R_0805_2012Metric	1
R4	1.01M 1%	R_0805_2012Metric	1
R5	98.9 0.1%	R_0805_2012Metric	1
R6	993.6 0.1%	R_0805_2012Metric	1
R7	98.6k 0.1%	R_0805_2012Metric	1
R8	4.7M 0.1%	R_0805_2012Metric	1
R9	47	R_2512_6332Metric	1
R10	100	R_2512_6332Metric	1
R11	1	R_0805_2012Metric	1
R12	4.8M	R_0805_2012Metric	1
R13, R30, R31, R40	100k	R_0805_2012Metric	4
R14, R18, R20, R25, R26, R32, R34, R36	1k	R_0805_2012Metric	8
R15, R19, R21, R27, R28, R29, R35	10k	R_0805_2012Metric	7
R16, R33, R38	10M 0.1%	R_0805_2012Metric	3
R17	7.15k	R_0805_2012Metric	1
R22, R23, R24, R41, R43	330	R_0805_2012Metric	5
R39	1M	R_0805_2012Metric	1
U1	LMC6482	SOIC-8_3.9x4.9mm_P1.27mm	1
U2	74LVC1G34	SOT-23-5	1
U3, U4	MAX4619	SOIC-16_3.9x9.9mm_P1.27mm	2
U5, U6, U7, U9, U10	AQY212	DIP-4_W7.62mm	5

2. Route A: soldering onto the PCB

1. Inspect the bare board. Check for shorts with a multimeter: +3V3 to GND (J4-3 to J4-1) must **not** beep.
2. Solder in this order (lowest parts first, so the board sits flat): **(a)** SOIC chips U1, U3, U4, then U2 · **(b)** SOT-23 parts Q1–Q3, D1–D3 · **(c)** SOD-123 diodes D4–D6 · **(d)** 0805 resistors and capacitors · **(e)** 2512 resistors R9, R10 · **(f)** AQY212EH SSRs U5–U7, U9, U10 · **(g)** relays K1, K2 · **(h)** C1 · **(i)** J5, J6 terminals · **(j)** sockets J1–J4, J7 and header J8.
3. Use the placement map below and match every polarised part to its silkscreen mark (table 2.1).
4. Wash off the flux around J5, U1, R16/C4, R38/R39 and C1 with isopropyl alcohol and let it dry; flux residue leaks current at the 10 MΩ – 1 GΩ level. Do not conformal-coat across those areas.
5. Tick every part on the BOM table as you fit it.

2.1 Orientation

Parts	How to orient
U1, U3, U4 (SOIC)	Pin 1 = dot / bevelled edge on the chip ↔ the dot or "1" on the silkscreen
U2 (SOT-23-5)	Pins 1–3 on one side, 4–5 on the other; match the silkscreen outline
Q1, Q2, Q3, D1, D2, D3 (SOT-23)	Single pin (pin 3) on one side, two pins on the other; match the outline
D4, D5, D6 (SOD-123)	Cathode band = pad 1. D4/D5 cathode to +3V3 (flyback); D6 cathode to FH
U5, U6, U7, U9, U10 (AQY212EH)	Pin 1 = dot/notch = LED anode
K1, K2 (G6S-2)	Asymmetric pin pattern only fits one way; coil = pins 1 and 12
C1 (bipolar)	Either way round. Do not use a polarised capacitor here
J5 (DUT)	Pin 1 = FH (top). Order FH, SH, SL, FL. 2-wire parts: link 1–2 and 3–4
J1–J4 (module sockets)	J1/J2 pin 1 at the right end, J3/J4 pin 1 at the left; the WeAct module plugs in USB-C facing the right edge

2.2 Placement map

See **Appendix A** (landscape page after the wire list): top side of the PCB with every reference designator.

3. Route B: hand-wired prototype

- **Use perfboard (2.54 mm), not a solderless breadboard.** A breadboard's contact leakage and stray capacitance ruin the TIA (≥ 100 MΩ) and pF ranges; perfboard is acceptable for a prototype. Expect the 1 GΩ range to be worse than on the PCB.
- Lay the parts out as in the wiring map: it is the PCB placement, so related parts are already next to each other.
- Run a **GND rail** and a **+3V3 rail** (tinned bus wire) along the board first. Pins drawn black go to GND, red to +3V3 (rail list at the end).
- Then solder the numbered wires from the wire list, ticking each one. Each wire connects two pins of the same net; the list is the minimum set of wires that joins every net, so do not add extras.
- The 44 wires marked ⚠ **short** carry sensitive analog signals (FH, FL, SH, SL, SH_DIV, VCAP, C_BLOCK, RF_TOP, VN_SENSE, MUXR_COM, VBIAS). Keep them short, away from the relay coils, and do not bundle them with the digital wires.
- Show one block at a time in draw.io: *Extras* ▸ *Layers* (Ctrl+Shift+L), or use the per-block pages of the PDF.

3.1 Through-hole substitutes and adapters

Parts	On the PCB	For perfboard

U1	LMC6482AIM/NOPB, SOIC-8	SOIC-8 → DIP-8 adapter board (keep the SOIC part: 20 fA input current is needed for the TIA)
U3, U4	MAX4619CSE+, SOIC-16	SOIC-16 → DIP-16 adapter, or the DIP-16 version MAX4619CPE+ (same pinout)
U2	74LVC1G34, SOT-23-5	SOT-23-5 → DIP adapter
Q1	AO3401A P-MOSFET, SOT-23	SOT-23 → DIP adapter (pins 1 G, 2 S, 3 D)
Q2, Q3	BC847B NPN, SOT-23	BC547B (TO-92). Pin order differs: TO-92 flat face towards you, left→right = C, B, E
D1, D2	BAV199 low-leakage pair, SOT-23	SOT-23 → DIP adapter (do not substitute 1N4148s here: leakage matters on VCAP and VN_SENSE)
D3	BAT54S Schottky pair, SOT-23	SOT-23 → DIP adapter, or two BAT54 / 1N5819 diodes (pin 3 = middle)
D4–D6	1N4148W, SOD-123	1N4148 (DO-35). Band = cathode = pin 1
0805 resistors	R1–R8, R12–R43	¼ W metal-film, same value and tolerance (0.1 % parts for R5–R8, R16, R33, R38)
0805 capacitors	C2–C21	Ceramic, same value: C0G/NP0 for 10 pF, 100 pF, 1 nF; X7R for 68 nF, 100 nF; 10 µF X5R or tantalum
R9, R10	2512	R9 47 Ω and R10 100 Ω, 2 W through-hole
R11	1 Ω 0805 (pulse)	1 Ω, ½ W through-hole
U5–U7, U9, U10	AQY212EH, DIP-4	unchanged (already through-hole)
K1, K2	G6S-2 DC3	unchanged (through-hole, pins on a 2.54 mm grid)
C1	1000 µF 35 V bipolar	unchanged (radial)
J1–J4, J7, J8, J5, J6	headers / terminals	unchanged; J5/J6 5.08 mm terminals fit 2.54 mm perfboard

3.2 Wiring map

See **Appendix B** (the last six A3 pages): an overview with every wire, then one page per block with only that block's wires highlighted and listed. The same drawing is editable in **ESWCap-wiring-map.drawio**.

4. Bring-up tests (both routes)

1. **Continuity, no power:** for every wire in the list, beep from the "From" pin to the "To" pin. +3V3↔GND must not beep.
2. **Power without the module:** feed 3.3 V from a bench supply into J4-3/4 (+) and J4-1/2 (−), current limit 150 mA. Expect a few mA. VREF (U4 side, R17/R18 junction) ≈ 0.405 V, VBIAS (R19/R21 junction) ≈ 1.65 V.
3. **Relays without the module:** put 3.3 V through 1 kΩ onto J1-2 (K1_DRV): K1 must click; FL (J5-4) must no longer beep to GND. Same for J2-19 (K2_DRV) and K2. Never energise K2 without K1 in normal use.
4. **SSRs without the module:** 3.3 V on J2-20 (SSR13) must make U5 and U6 conduct (V_BIAS↔FH through 47 Ω, VCAP↔GND through 1 Ω); 3.3 V on J1-19 must make U7 conduct (FH↔GND through 100 Ω).
5. **Module:** only after the firmware with the F1 pin-map update is flashed. Plug in, USB power, open the console on J7 (115200 8N1): expect the banner and `SYSCLK 170000000 Hz`.
6. **First measurements:** C-mode with an empty socket (`zero`), then known capacitors; R-mode only

with the combined firmware build, starting with a 1 kΩ resistor (V at FL/MUXR_COM ≈ 0.405 V).
7. **V_BIAS last:** 5 V, then 10 V, then 20 V on J6, with J5 covered.

5. Wire list (124 wires)

✓	#	Block	Net	From	To	PCB mm	Note
☐	1	Power, relays & control	BIAS_EN	J1-10	R35-1	53.9	
☐	2	Power, relays & control	BIAS_EN	R35-1	J8-2	15.7	
☐	3	Power, relays & control	BIAS_MON	J1-15	R36-1	54.9	
☐	4	Power, relays & control	BIAS_MON_F	C17-1	R36-2	2.1	
☐	5	Power, relays & control	BIAS_MON_F	R36-2	J8-3	8.4	
☐	6	Power, relays & control	BIAS_SET	J2-10	R34-1	68.9	
☐	7	Power, relays & control	BIAS_SET_F	C16-1	R34-2	2.1	
☐	8	Power, relays & control	BIAS_SET_F	R34-2	J8-1	6.7	
☐	9	Power, relays & control	BIAS_UART_RX	J1-8	J8-7	59.3	
☐	10	Power, relays & control	BIAS_UART_TX	J1-9	J8-6	59.3	
☐	11	Power, relays & control	K1_BASE	Q2-1	R27-1	4.4	
☐	12	Power, relays & control	K1_BASE	R27-1	R25-2	3.3	
☐	13	Power, relays & control	K1_C	D4-2	Q2-3	4.4	
☐	14	Power, relays & control	K1_C	Q2-3	R41-1	11.4	
☐	15	Power, relays & control	K1_C	R41-1	K1-12	8.5	
☐	16	Power, relays & control	K1_DRV	J1-2	R25-1	80.7	
☐	17	Power, relays & control	K2_BASE	Q3-1	R28-1	4.4	
☐	18	Power, relays & control	K2_BASE	R28-1	R26-2	3.3	
☐	19	Power, relays & control	K2_C	D5-2	Q3-3	4.4	

☐	20	Power, relays & control	K2_C	Q3-3	K2-12	8.6	
☐	21	Power, relays & control	K2_DRV	J2-19	R26-1	54.9	
☐	22	Power, relays & control	UART_RX	J1-13	J7-2	84.7	
☐	23	Power, relays & control	UART_TX	J1-14	J7-1	84.7	
☐	24	C-mode	C_BLOCK	C1-2	C2-2	10.2	⚠ short
☐	25	C-mode	C_BLOCK	C2-2	R33-2	6.0	⚠ short
☐	26	C-mode	C_BLOCK	C1-2	K1-10	25.1	⚠ short
☐	27	C-mode	C_CH0	R3-1	U3-12	9.3	
☐	28	C-mode	C_CH1	R2-1	U3-13	8.5	
☐	29	C-mode	C_CH2	R1-1	U3-2	3.5	
☐	30	C-mode	C_CH3	R4-1	U3-1	13.1	
☐	31	C-mode	C_MUX_COM	U2-4	U3-4	12.0	
☐	32	C-mode	DRIVE	J2-6	U2-2	51.6	
☐	33	C-mode	DRIVE	U2-2	R29-1	4.0	
☐	34	C-mode	MUXC_A0	J1-7	U3-10	52.8	
☐	35	C-mode	MUXC_A0	U3-10	U3-11	1.3	
☐	36	C-mode	MUXC_A1	J1-6	U3-9	55.3	
☐	37	C-mode	MUXC_EN	J1-5	R30-1	53.8	
☐	38	C-mode	MUXC_EN	R30-1	U3-6	9.3	
☐	39	C-mode	MUXC_P	U3-14	U3-5	5.6	
☐	40	C-mode	MUXC_Q	U3-15	U3-3	5.1	
☐	41	C-mode	VCAP	C1-1	R3-2	6.6	⚠ short
☐	42	C-mode	VCAP	R3-2	R4-2	3.6	⚠ short
☐	43	C-mode	VCAP	R3-2	R2-2	3.6	⚠ short
☐	44	C-mode	VCAP	R2-2	R1-2	3.6	⚠ short
☐	45	C-mode	VCAP	R4-2	D1-3	5.8	⚠ short
☐	46	C-mode	VCAP	D1-3	R32-1	2.1	⚠ short

☐	47	C-mode	VCAP	C1-1	R33-1	11.4	⚠ short
☐	48	C-mode	VCAP	R33-1	C2-1	6.0	⚠ short
☐	49	C-mode	VCAP	R32-1	U6-4	13.1	⚠ short
☐	50	C-mode	VCAP_ADC	J2-7	R32-2	59.8	
☐	51	Bias & safety	C_MODE_SW	J2-14	R43-1	78.9	
☐	52	Bias & safety	FH	D6-1	K1-9	7.3	⚠ short
☐	53	Bias & safety	FH	D6-1	J5-1	13.0	⚠ short
☐	54	Bias & safety	FH	K1-9	R12-2	22.3	⚠ short
☐	55	Bias & safety	FH	R12-2	R9-2	17.7	⚠ short
☐	56	Bias & safety	FH	R9-2	U7-4	15.8	⚠ short
☐	57	Bias & safety	SH_DIV	C20-1	R39-1	3.0	⚠ short
☐	58	Bias & safety	SH_DIV	R39-1	R38-2	6.8	⚠ short
☐	59	Bias & safety	SH_DIV	R38-2	C21-2	3.0	⚠ short
☐	60	Bias & safety	SH_DIV	C20-1	U10-3	7.3	⚠ short
☐	61	Bias & safety	SSR13	J2-20	R23-1	23.0	
☐	62	Bias & safety	SSR13	R23-1	R22-1	58.5	
☐	63	Bias & safety	SSR1_LED	R22-2	U5-1	4.7	
☐	64	Bias & safety	SSR1_SW	R9-1	U5-3	9.3	
☐	65	Bias & safety	SSR2_CTRL	J1-19	R24-1	51.5	
☐	66	Bias & safety	SSR2_LED	R24-2	U7-1	11.0	
☐	67	Bias & safety	SSR2_SW	R10-1	U7-3	9.3	
☐	68	Bias & safety	SSR3_LED	R23-2	U6-1	13.0	
☐	69	Bias & safety	SSR3_SW	R11-1	U6-3	3.8	
☐	70	Bias & safety	U10_LED_K	R43-2	U10-2	11.8	
☐	71	Bias & safety	V_BIAS	J6-1	R12-1	6.2	
☐	72	Bias & safety	V_BIAS	R12-1	U5-4	12.2	
☐	73	R-mode Kelvin	MUXR_A0	J1-4	U4-11	42.0	
☐	74	R-mode Kelvin	MUXR_A0	U4-11	U4-10	1.3	

☐	75	R-mode Kelvin	MUXR_A1	J1-3	U4-9	45.5	
☐	76	R-mode Kelvin	MUXR_COM	J2-13	U4-4	37.4	⚠ short
☐	77	R-mode Kelvin	MUXR_COM	U4-4	K2-5	18.5	⚠ short
☐	78	R-mode Kelvin	MUXR_EN	J1-17	U4-6	21.3	
☐	79	R-mode Kelvin	MUXR_EN	U4-6	R31-1	6.3	
☐	80	R-mode Kelvin	MUXR_P	U4-14	U4-5	5.6	
☐	81	R-mode Kelvin	MUXR_Q	U4-15	U4-3	5.1	
☐	82	R-mode Kelvin	OP1_OUT	C3-1	J2-8	25.0	
☐	83	R-mode Kelvin	OP1_OUT	C3-1	R14-1	54.0	
☐	84	R-mode Kelvin	Q1_D	D6-2	Q1-3	3.3	
☐	85	R-mode Kelvin	Q1_GATE	Q1-1	R13-2	9.4	
☐	86	R-mode Kelvin	Q1_GATE	R13-2	R14-2	2.5	
☐	87	R-mode Kelvin	Q1_GATE	R13-2	R15-2	4.5	
☐	88	R-mode Kelvin	R_CH0	R5-1	U4-12	5.3	
☐	89	R-mode Kelvin	R_CH1	R6-1	U4-13	2.6	
☐	90	R-mode Kelvin	R_CH2	R7-1	U4-2	9.2	
☐	91	R-mode Kelvin	R_CH3	R8-1	U4-1	12.1	
☐	92	R-mode Kelvin	R_FORCE	J1-18	R15-1	42.1	
☐	93	R-mode Kelvin	SH	C21-1	R38-1	3.0	⚠ short
☐	94	R-mode Kelvin	SH	R38-1	J5-2	7.7	⚠ short
☐	95	R-mode Kelvin	SH	C21-1	U9-3	18.8	⚠ short
☐	96	R-mode Kelvin	SL	J5-3	R40-1	39.3	⚠ short
☐	97	R-mode Kelvin	SL_BUF	R40-2	U1-5	3.9	⚠ short
☐	98	R-mode Kelvin	U9_LED_K	R41-2	U9-2	12.6	
☐	99	R-mode Kelvin	VBUF	J1-20	J2-17	17.0	
☐	100	R-mode Kelvin	VBUF	J1-20	U1-7	22.8	
☐	101	R-mode Kelvin	VBUF	U1-7	U1-6	1.3	
☐	102	R-mode Kelvin	VN_IN	R20-1	U9-4	3.1	⚠ short
☐	103	R-mode Kelvin	VN_SENSE	C7-1	R20-2	2.1	⚠ short

☐	104	R-mode Kelvin	VN_SENSE	C7-1	D2-3	4.0	⚠ short
☐	105	R-mode Kelvin	VN_SENSE	R20-2	U10-4	9.3	⚠ short
☐	106	R-mode Kelvin	VN_SENSE	U10-4	J2-15	64.6	⚠ short
☐	107	R-mode Kelvin	VREF	C3-2	R17-2	3.0	
☐	108	R-mode Kelvin	VREF	R17-2	R18-1	3.5	
☐	109	R-mode Kelvin	VREF	R18-1	C5-1	3.0	
☐	110	R-mode Kelvin	VREF	R17-2	J2-9	21.8	
☐	111	R-mode TIA	FL	C4-1	R16-1	2.5	⚠ short
☐	112	R-mode TIA	FL	C4-1	U1-2	4.0	⚠ short
☐	113	R-mode TIA	FL	R16-1	K2-4	4.5	⚠ short
☐	114	R-mode TIA	FL	K2-4	K1-4	11.0	⚠ short
☐	115	R-mode TIA	FL	K1-4	J5-4	22.8	⚠ short
☐	116	R-mode TIA	RF_TOP	C4-2	R16-2	2.5	⚠ short
☐	117	R-mode TIA	RF_TOP	R16-2	K2-10	11.1	⚠ short
☐	118	R-mode TIA	TIA_OUT	D3-3	U1-1	11.2	
☐	119	R-mode TIA	TIA_OUT	U1-1	K2-9	10.6	
☐	120	R-mode TIA	TIA_OUT	U1-1	J2-12	49.2	
☐	121	R-mode TIA	VBIAS	C6-1	R21-1	2.5	⚠ short
☐	122	R-mode TIA	VBIAS	R21-1	R19-2	3.1	⚠ short
☐	123	R-mode TIA	VBIAS	R19-2	U1-3	11.3	⚠ short
☐	124	R-mode TIA	VBIAS	U1-3	J2-11	52.8	⚠ short

Rail connections (82)

✓	Pin	Rail
☐	C10-2	GND
☐	C11-2	GND
☐	C12-2	GND
☐	C13-2	GND
☐	C14-2	GND
☐	C15-2	GND

- [ ] C16-2 GND
- [ ] C17-2 GND
- [ ] C20-2 GND
- [ ] C5-2 GND
- [ ] C6-2 GND
- [ ] C7-2 GND
- [ ] C8-2 GND
- [ ] C9-2 GND
- [ ] D1-1 GND
- [ ] D2-1 GND
- [ ] D3-1 GND
- [ ] J1-1 GND
- [ ] J2-1 GND
- [ ] J3-1 GND
- [ ] J3-2 GND
- [ ] J4-1 GND
- [ ] J4-2 GND
- [ ] J6-2 GND
- [ ] J7-3 GND
- [ ] J8-4 GND
- [ ] J8-8 GND
- [ ] K1-3 GND
- [ ] Q2-2 GND
- [ ] Q3-2 GND
- [ ] R10-2 GND
- [ ] R11-2 GND
- [ ] R18-2 GND
- [ ] R21-2 GND
- [ ] R27-2 GND

☐ R28-2 GND

☐ R29-2 GND

☐ R35-2 GND

☐ R39-2 GND

☐ R5-2 GND

☐ R6-2 GND

☐ R7-2 GND

☐ R8-2 GND

☐ U1-4 GND

☐ U2-3 GND

☐ U3-8 GND

☐ U4-8 GND

☐ U5-2 GND

☐ U6-2 GND

☐ U7-2 GND

☐ C10-1 +3V3

☐ C11-1 +3V3

☐ C12-1 +3V3

☐ C13-1 +3V3

☐ C14-1 +3V3

☐ C15-1 +3V3

☐ C8-1 +3V3

☐ C9-1 +3V3

☐ D1-2 +3V3

☐ D2-2 +3V3

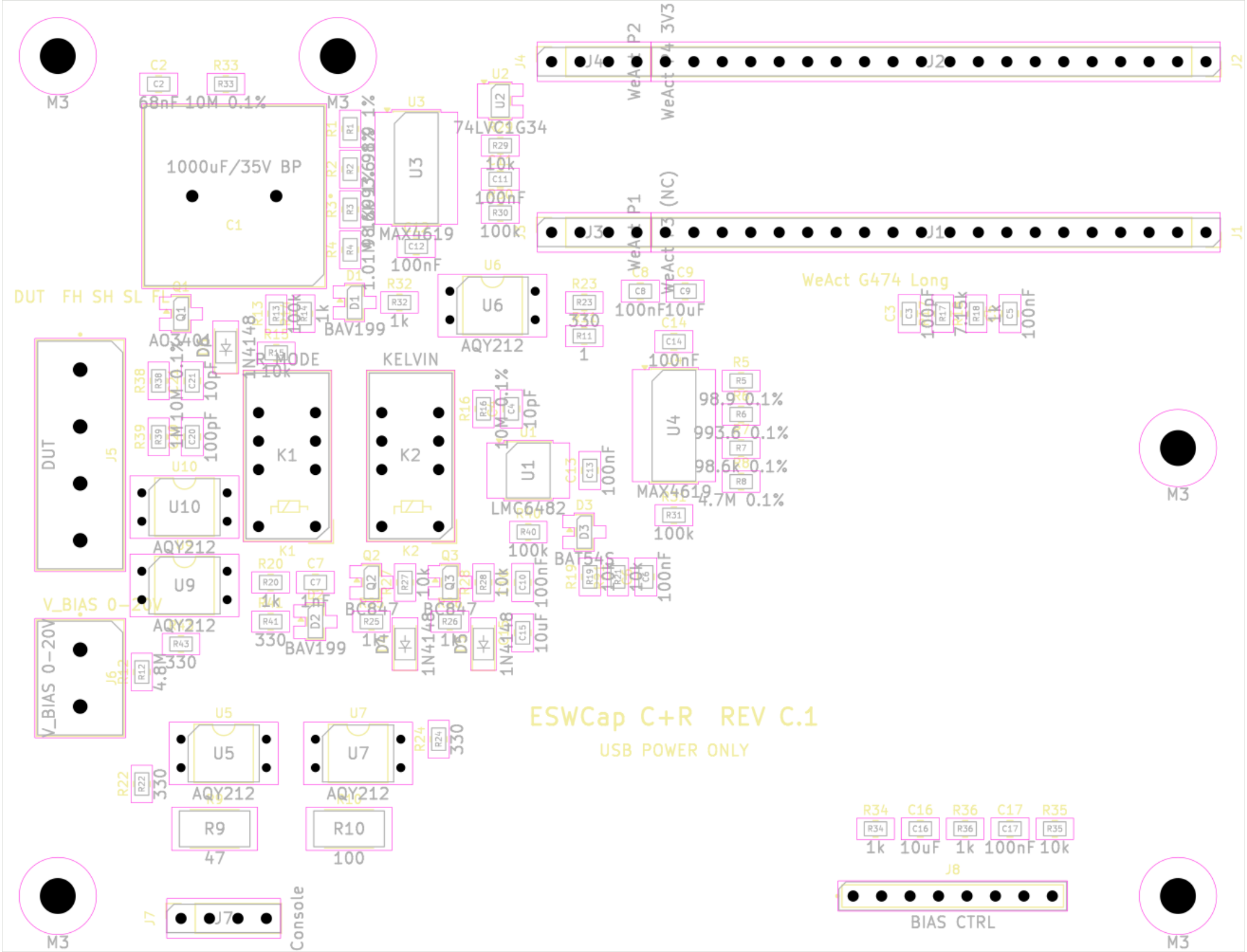
☐ D3-2 +3V3

☐ D4-1 +3V3

☐ D5-1 +3V3

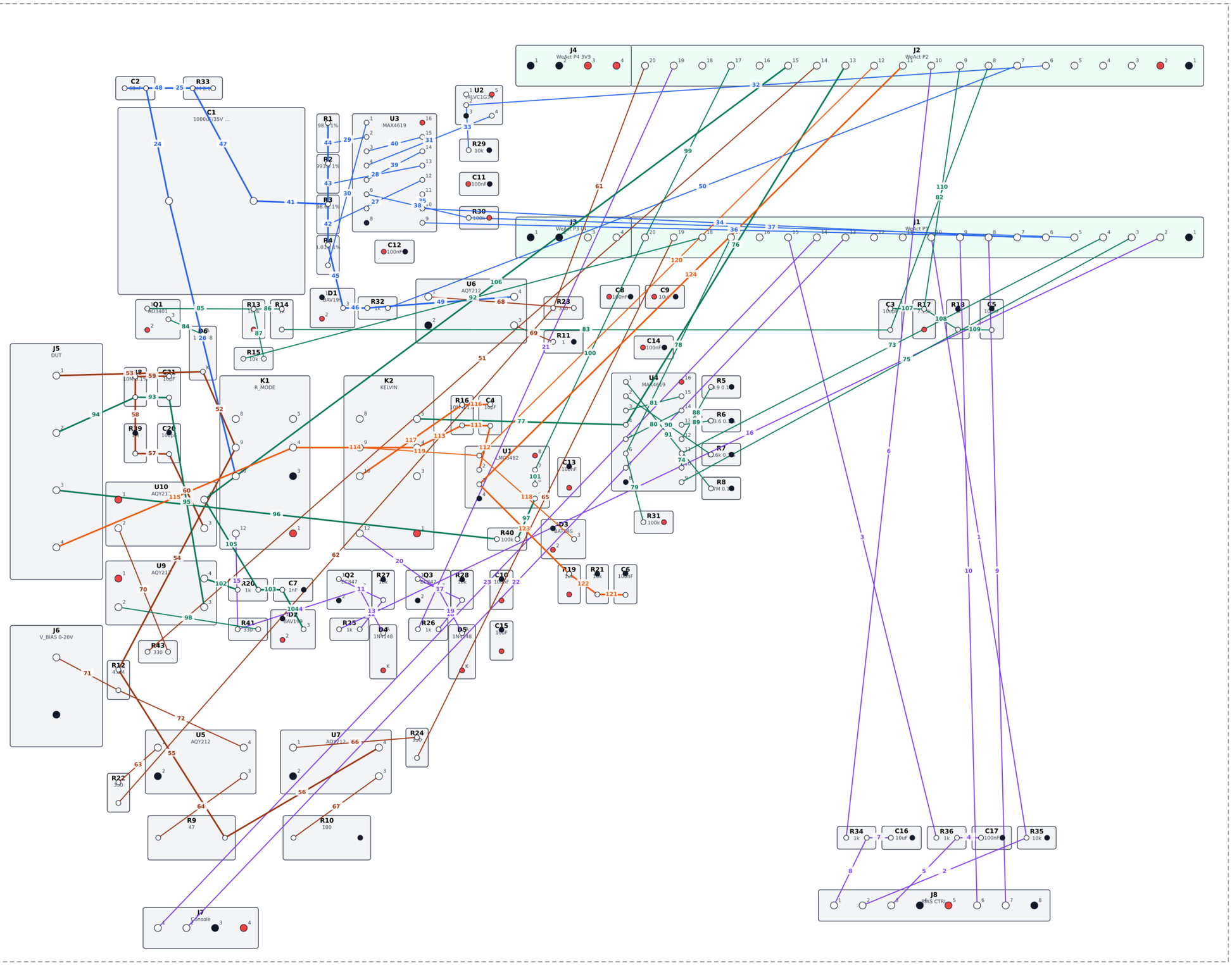
☐ J2-2 +3V3

- [ ] J4-3 +3V3
- [ ] J4-4 +3V3
- [ ] J7-4 +3V3
- [ ] J8-5 +3V3
- [ ] K1-1 +3V3
- [ ] K2-1 +3V3
- [ ] Q1-2 +3V3
- [ ] R13-1 +3V3
- [ ] R17-1 +3V3
- [ ] R19-1 +3V3
- [ ] R30-2 +3V3
- [ ] R31-2 +3V3
- [ ] U1-8 +3V3
- [ ] U10-1 +3V3
- [ ] U2-5 +3V3
- [ ] U3-16 +3V3
- [ ] U4-16 +3V3
- [ ] U9-1 +3V3


ESWCap C+R REV C.1
USB POWER ONLY
WeAct G474 Long
1000uF/35V BP
MAX4619
74LVC1G34
AQY212
LMC6482
BC847
BAT54S
BAV199
1N4148
AO3401
DUT
KELVIN
R_MODE
V_BIAS 0-20V
BIAS CTRL
Console
M3

ESWCap Rev C.1 — wiring map (all blocks)

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

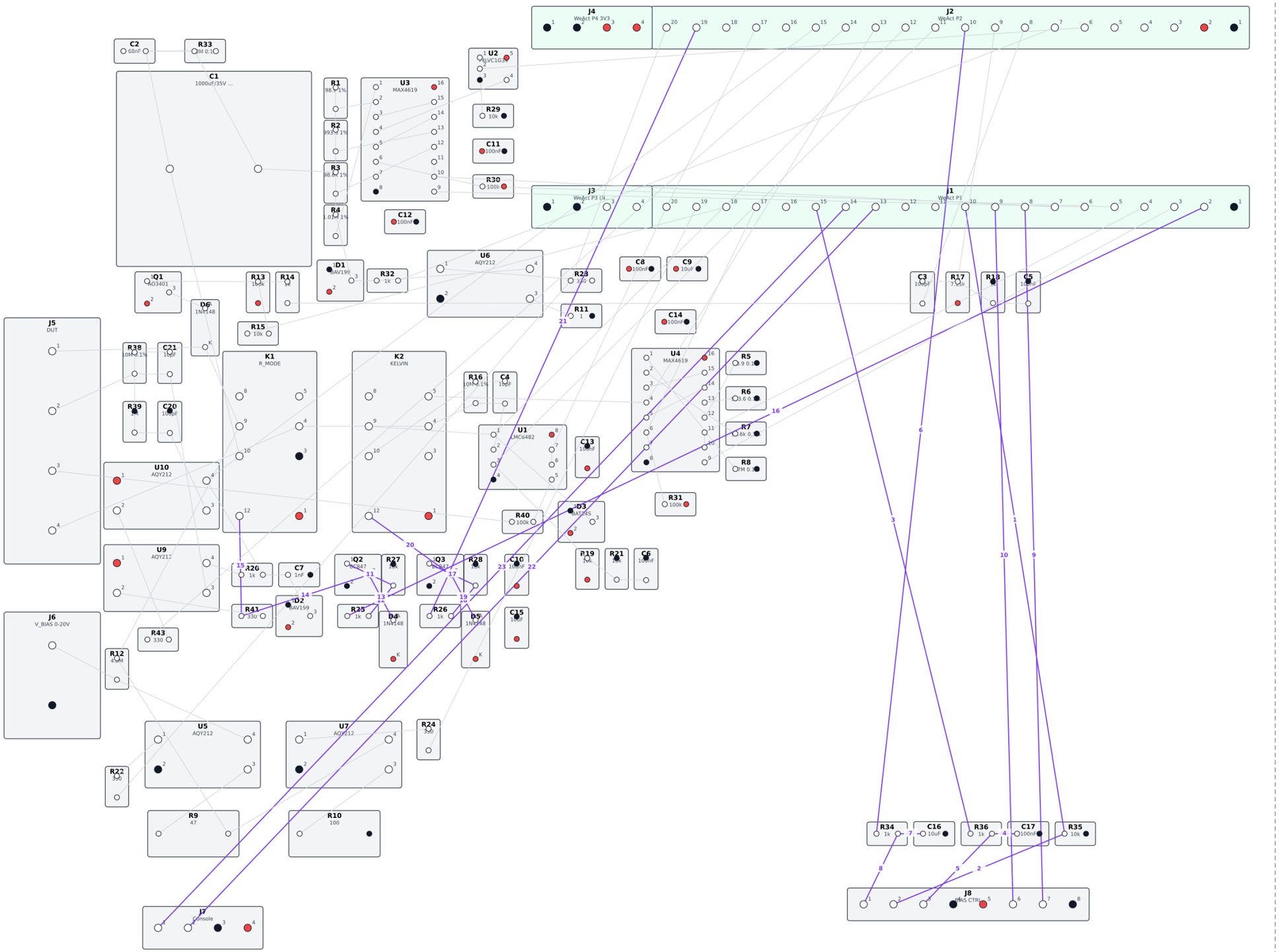
● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

wire numbers = ESWCap-wire-list.csv

ESWCap Rev C.1 — wiring map: Power, relays & control

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

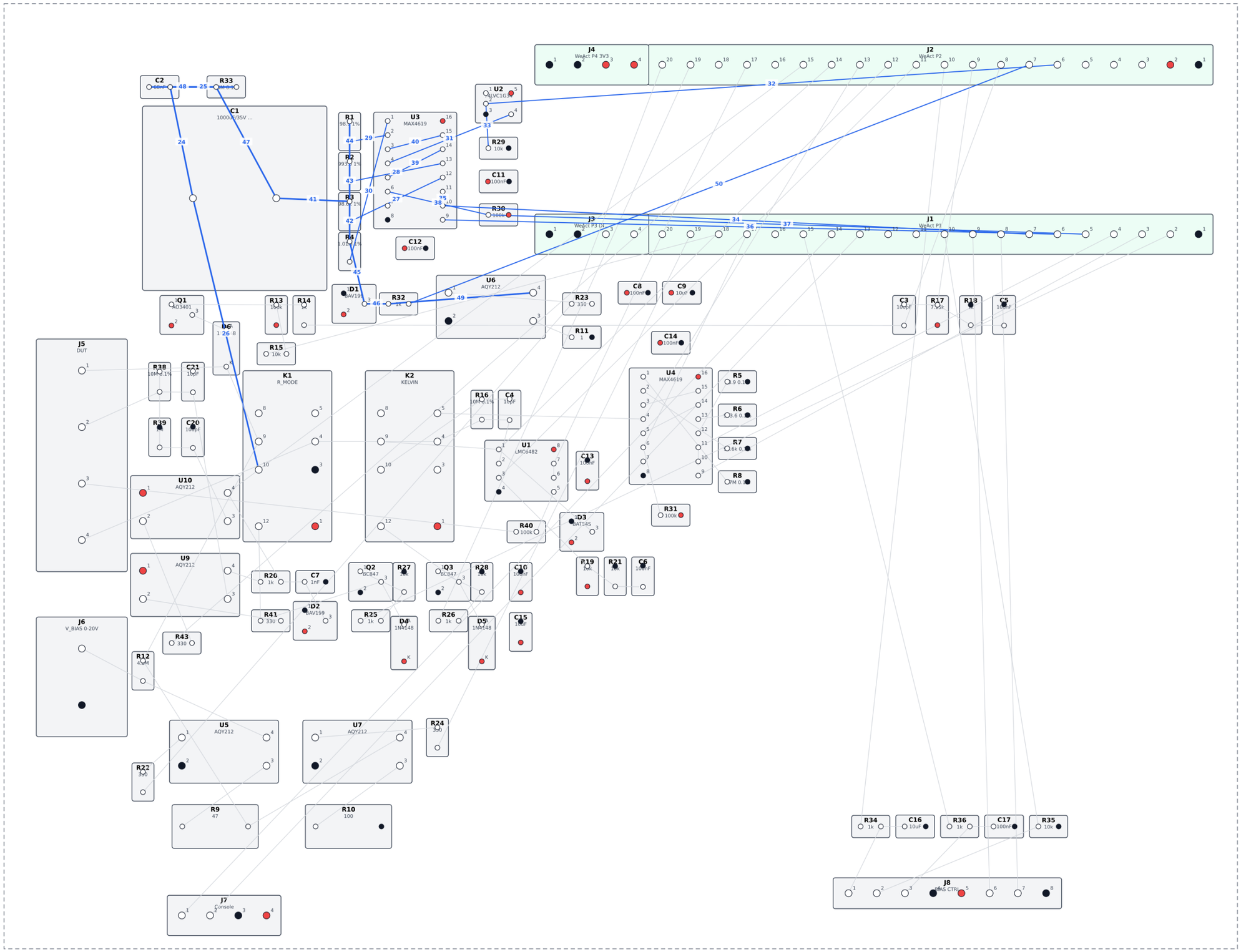
wire numbers = ESWCap-wire-list.csv

Wires on this page (Power, relays & control):

1 J1-10 → R35-1 (BIAS_EN)
2 R35-1 → J8-2 (BIAS_EN)
3 J1-15 → R36-1 (BIAS_MON)
4 C17-1 → R36-2 (BIAS_MON_F)
5 R36-2 → J8-3 (BIAS_MON_F)
6 J2-10 → R34-1 (BIAS_SET)
7 C16-1 → R34-2 (BIAS_SET_F)
8 R34-2 → J8-1 (BIAS_SET_F)
9 J1-8 → J8-7 (BIAS_UART_RX)
10 J1-9 → J8-6 (BIAS_UART_TX)
11 Q2-1 → R27-1 (K1_BASE)
12 R27-1 → R25-2 (K1_BASE)
13 D4-2 → Q2-3 (K1_C)
14 Q2-3 → R41-1 (K1_C)
15 R41-1 → K1-12 (K1_C)
16 J1-2 → R25-1 (K1_DRV)
17 Q3-1 → R28-1 (K2_BASE)
18 R28-1 → R26-2 (K2_BASE)
19 D5-2 → Q3-3 (K2_C)
20 Q3-3 → K2-12 (K2_C)
21 J2-19 → R26-1 (K2_DRV)
22 J1-13 → J7-2 (UART_RX)
23 J1-14 → J7-1 (UART_TX)

ESWCap Rev C.1 — wiring map: C-mode

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

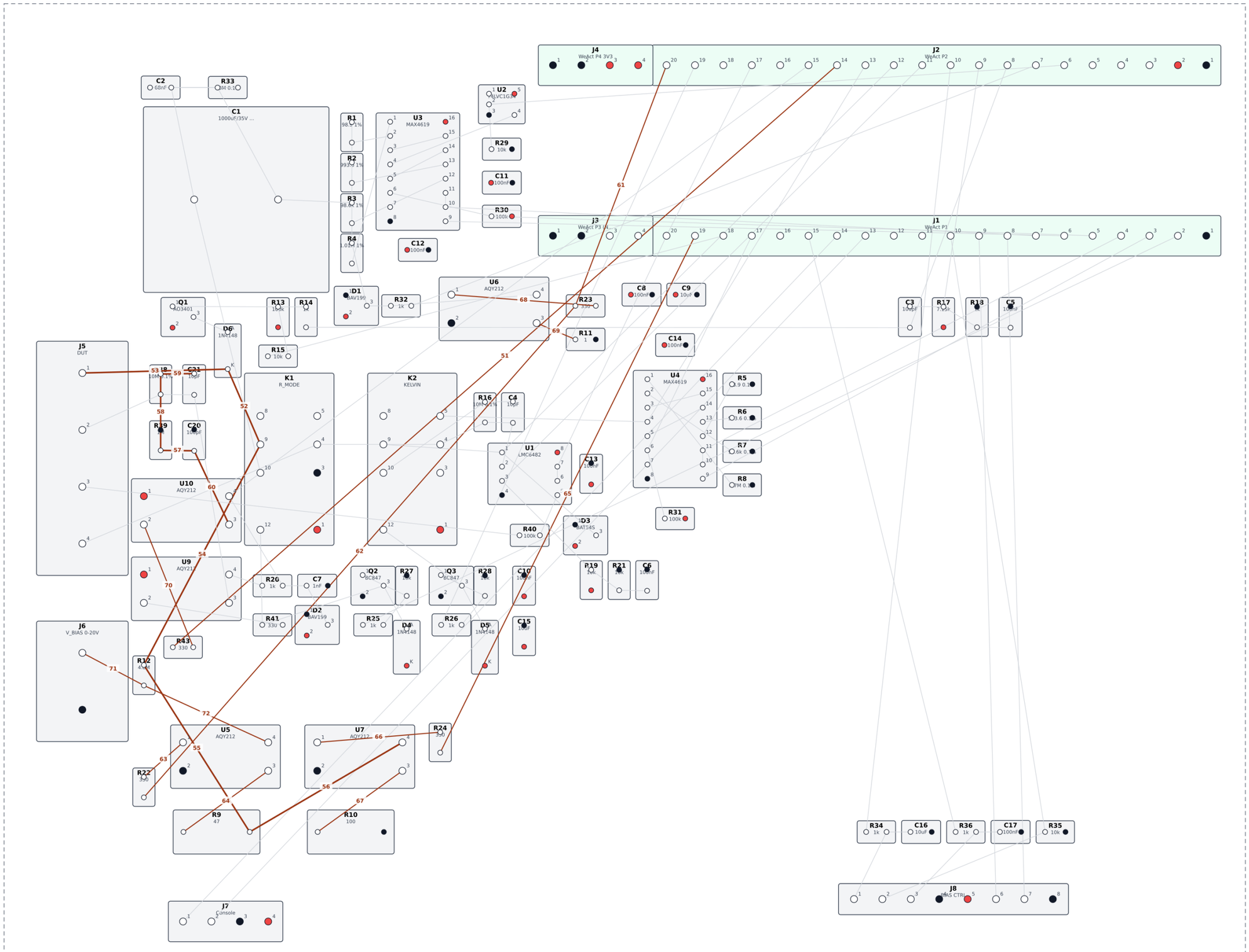
wire numbers = ESWCap-wire-list.csv

Wires on this page (C-mode):

24 C1-2 → C2-2 (C_BLOCK) ⚠
25 C2-2 → R33-2 (C_BLOCK) ⚠
26 C1-2 → K1-10 (C_BLOCK) ⚠
27 R3-1 → U3-12 (C_CH0)
28 R2-1 → U3-13 (C_CH1)
29 R1-1 → U3-2 (C_CH2)
30 R4-1 → U3-1 (C_CH3)
31 U2-4 → U3-4 (C_MUX_COM)
32 J2-6 → U2-2 (DRIVE)
33 U2-2 → R29-1 (DRIVE)
34 J1-7 → U3-10 (MUXC_A0)
35 U3-10 → U3-11 (MUXC_A0)
36 J1-6 → U3-9 (MUXC_A1)
37 J1-5 → R30-1 (MUXC_EN)
38 R30-1 → U3-6 (MUXC_EN)
39 U3-14 → U3-5 (MUXC_P)
40 U3-15 → U3-3 (MUXC_Q)
41 C1-1 → R3-2 (VCAP) ⚠
42 R3-2 → R4-2 (VCAP) ⚠
43 R3-2 → R2-2 (VCAP) ⚠
44 R2-2 → R1-2 (VCAP) ⚠
45 R4-2 → D1-3 (VCAP) ⚠
46 D1-3 → R32-1 (VCAP) ⚠
47 C1-1 → R33-1 (VCAP) ⚠
48 R33-1 → C2-1 (VCAP) ⚠
49 R32-1 → U6-4 (VCAP) ⚠
50 J2-7 → R32-2 (VCAP_ADC)

ESWCap Rev C.1 — wiring map: Bias & safety

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

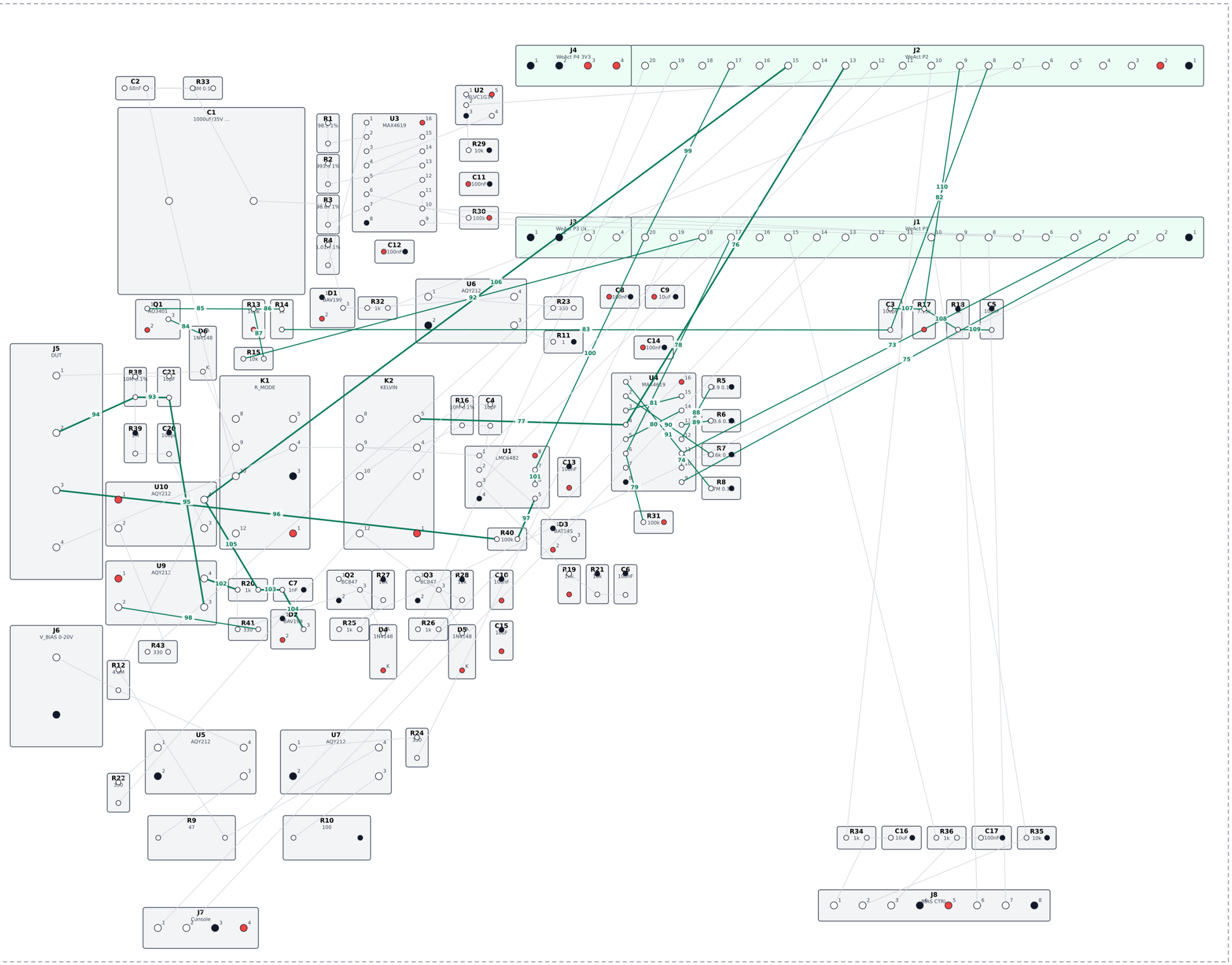
wire numbers = ESWCap-wire-list.csv

Wires on this page (Bias & safety):

51 J2-14 → R43-1 (C_MODE_SW)
52 D6-1 → K1-9 (FH) ⚠
53 D6-1 → J5-1 (FH) ⚠
54 K1-9 → R12-2 (FH) ⚠
55 R12-2 → R9-2 (FH) ⚠
56 R9-2 → U7-4 (FH) ⚠
57 C20-1 → R39-1 (SH_DIV) ⚠
58 R39-1 → R38-2 (SH_DIV) ⚠
59 R38-2 → C21-2 (SH_DIV) ⚠
60 C20-1 → U10-3 (SH_DIV) ⚠
61 J2-20 → R23-1 (SSR13)
62 R23-1 → R22-1 (SSR13)
63 R22-2 → U5-1 (SSR1_LED)
64 R9-1 → U5-3 (SSR1_SW)
65 J1-19 → R24-1 (SSR2_CTRL)
66 R24-2 → U7-1 (SSR2_LED)
67 R10-1 → U7-3 (SSR2_SW)
68 R23-2 → U6-1 (SSR3_LED)
69 R11-1 → U6-3 (SSR3_SW)
70 R43-2 → U10-2 (U10_LED_K)
71 J6-1 → R12-1 (V_BIAS)
72 R12-1 → U5-4 (V_BIAS)

ESWCap Rev C.1 — wiring map: R-mode Kelvin

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

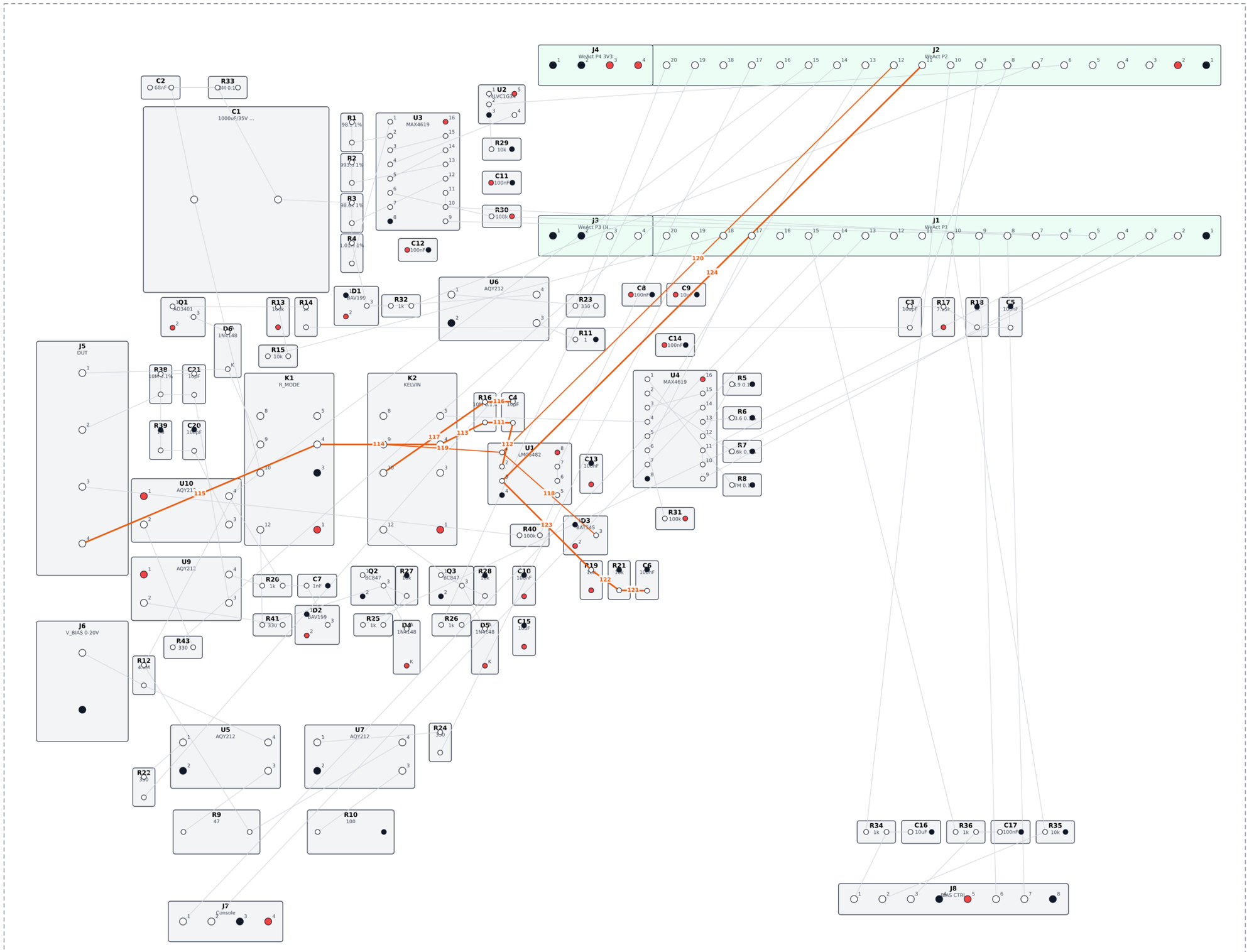
wire numbers = ESWCap-wire-list.csv

Wires on this page (R-mode Kelvin):

73 J1-4 → U4-11 (MUXR_A0)
74 U4-11 → U4-10 (MUXR_A0)
75 J1-3 → U4-9 (MUXR_A1)
76 J2-13 → U4-4 (MUXR_COM) ⚠
77 U4-4 → K2-5 (MUXR_COM) ⚠
78 J1-17 → U4-6 (MUXR_EN)
79 U4-6 → R31-1 (MUXR_EN)
80 U4-14 → U4-5 (MUXR_P)
81 U4-15 → U4-3 (MUXR_Q)
82 C3-1 → J2-8 (OP1_OUT)
83 C3-1 → R14-1 (OP1_OUT)
84 D6-2 → Q1-3 (Q1_D)
85 Q1-1 → R13-2 (Q1_GATE)
86 R13-2 → R14-2 (Q1_GATE)
87 R13-2 → R15-2 (Q1_GATE)
88 R5-1 → U4-12 (R_CH0)
89 R6-1 → U4-13 (R_CH1)
90 R7-1 → U4-2 (R_CH2)
91 R8-1 → U4-1 (R_CH3)
92 J1-18 → R15-1 (R_FORCE)
93 C21-1 → R38-1 (SH) ⚠
94 R38-1 → J5-2 (SH) ⚠
95 C21-1 → U9-3 (SH) ⚠
96 J5-3 → R40-1 (SL) ⚠
97 R40-2 → U1-5 (SL_BUF) ⚠
98 R41-2 → U9-2 (U9_LED_K)
99 J1-20 → J2-17 (VBUF)
100 J1-20 → U1-7 (VBUF)
101 U1-7 → U1-6 (VBUF)
102 R20-1 → U9-4 (VN_IN) ⚠
103 C7-1 → R20-2 (VN_SENSE) ⚠
104 C7-1 → D2-3 (VN_SENSE) ⚠
105 R20-2 → U10-4 (VN_SENSE) ⚠
106 U10-4 → J2-15 (VN_SENSE) ⚠
107 C3-2 → R17-2 (VREF)
108 R17-2 → R18-1 (VREF)
109 R18-1 → C5-1 (VREF)
110 R17-2 → J2-9 (VREF)

ESWCap Rev C.1 — wiring map: R-mode TIA

Parts are placed as on the PCB (top view). Each numbered line is one wire (ESWCap-wire-list.csv). Black pins → GND rail, red pins → +3V3 rail.



Legend

Power, relays & control wires

C-mode wires

Bias & safety wires

R-mode Kelvin wires

R-mode TIA wires

thick line = sensitive analog: keep short

● black pin = connect to GND rail

● red pin = connect to +3V3 rail

numbers on pins = pin numbers (K/A = diode)

wire numbers = ESWCap-wire-list.csv

Wires on this page (R-mode TIA):

111 C4-1 → R16-1 (FL) ⚠
112 C4-1 → U1-2 (FL) ⚠
113 R16-1 → K2-4 (FL) ⚠
114 K2-4 → K1-4 (FL) ⚠
115 K1-4 → J5-4 (FL) ⚠
116 C4-2 → R16-2 (RF_TOP) ⚠
117 R16-2 → K2-10 (RF_TOP) ⚠
118 D3-3 → U1-1 (TIA_OUT)
119 U1-1 → K2-9 (TIA_OUT)
120 U1-1 → J2-12 (TIA_OUT)
121 C6-1 → R21-1 (VBIAS) ⚠
122 R21-1 → R19-2 (VBIAS) ⚠
123 R19-2 → U1-3 (VBIAS) ⚠
124 U1-3 → J2-11 (VBIAS) ⚠